RWANDA HOUSE PRICE ESTIMATOR

Machine Learning and Streamlit Web Application

PROJECT REPORT

Prepared as a Machine Learning Project
2026

ABSTRACT

This project presents a Rwanda House Price Estimator, a machine-learning-based web application designed to estimate the price of a residential property from selected property characteristics. The application provides a simple interface through which a user enters property information and receives an estimated house price. The system is deployed using Streamlit, making the machine learning model accessible through a web browser. The project demonstrates how data science, machine learning, Python programming, and web application technologies can be combined to support property-price estimation in Rwanda.

1. INTRODUCTION

House prices can vary considerably depending on factors such as location, property size, number of bedrooms, number of bathrooms, land characteristics, and other housing attributes. Estimating a reasonable price manually can be difficult, particularly when many variables influence the market value. Machine learning provides an approach for learning relationships between historical housing data and property prices and then using those relationships to estimate prices for new properties.

The Rwanda House Price Estimator project applies this concept by developing a predictive model and connecting it to a Streamlit interface. The deployed application allows users to interact with the model without needing to write Python code.

2. PROBLEM STATEMENT

Property buyers, sellers, investors, and other users may have difficulty obtaining a quick and consistent estimate of a house's potential market price. Traditional estimation may depend heavily on manual comparison and individual judgement. There is therefore a need for a computerized system that can use available housing characteristics to provide an estimated property price quickly and consistently.

3. OBJECTIVES

3.1 General Objective

To design and implement a machine-learning-based system capable of estimating house prices in Rwanda from selected property characteristics.

3.2 Specific Objectives

- To collect and prepare housing data for machine learning.
- To identify relevant features that influence house prices.
- To train a machine learning model for house-price prediction.
- To evaluate the predictive performance of the trained model.
- To develop a user-friendly Streamlit web interface.
- To deploy the application so that users can access the estimator through a web browser.

4. METHODOLOGY

The project follows a typical machine learning workflow: data collection, data cleaning, exploratory analysis, feature preparation, model training, model evaluation, application development, and deployment.

4.1 Data Preparation

The housing dataset is prepared by checking the available variables, handling missing or invalid values where necessary, converting features into suitable numerical representations, and separating input variables from the target house-price variable.

4.2 Model Training

The prepared data is divided into training and testing portions. The machine learning algorithm is trained using the training data and evaluated using previously unseen testing data. The exact model and performance values should be reported from the final implementation and experiment results.

5. SYSTEM DESIGN

The system consists of three main components: the user interface, the machine learning prediction component, and the deployment environment.

Component	Function
User Interface	Collects property information from the user.
Machine Learning Model	Processes the input features and predicts the house price.
Streamlit Application	Connects the interface to the prediction model and displays the result.
Web Deployment	Makes the application accessible through a web browser.

6. IMPLEMENTATION

Python is used for data processing and machine learning. The Streamlit framework is used to build the interactive web interface. A typical prediction workflow is: the user enters property details, the application validates and transforms the inputs, the trained model generates a prediction, and the estimated price is displayed to the user.

The application is hosted on Streamlit's cloud platform and can be accessed using the project URL provided by the developer.

7. RESULTS AND TESTING

Testing should verify that the application accepts valid property information, produces a prediction, handles inappropriate or missing input appropriately, and displays the result correctly. Model evaluation should include the performance metrics obtained during testing, such as MAE, RMSE, or R^2 where applicable.

Because the deployed application's internal dataset and model metrics were not supplied with this report, specific accuracy values are intentionally not fabricated. They should be inserted from the actual model training and testing results.

8. ADVANTAGES

- Provides rapid house-price estimates.
- Easy to use through a web browser.
- Demonstrates practical application of machine learning.
- Can support preliminary property-price analysis.
- Can be improved as more reliable Rwanda housing data becomes available.

9. LIMITATIONS

- Prediction quality depends on the quality and representativeness of the training dataset.
- The estimated price is not necessarily the final market value of a property.
- Changes in the housing market can reduce model performance over time.
- Important variables not included in the dataset may affect the prediction.

10. RECOMMENDATIONS

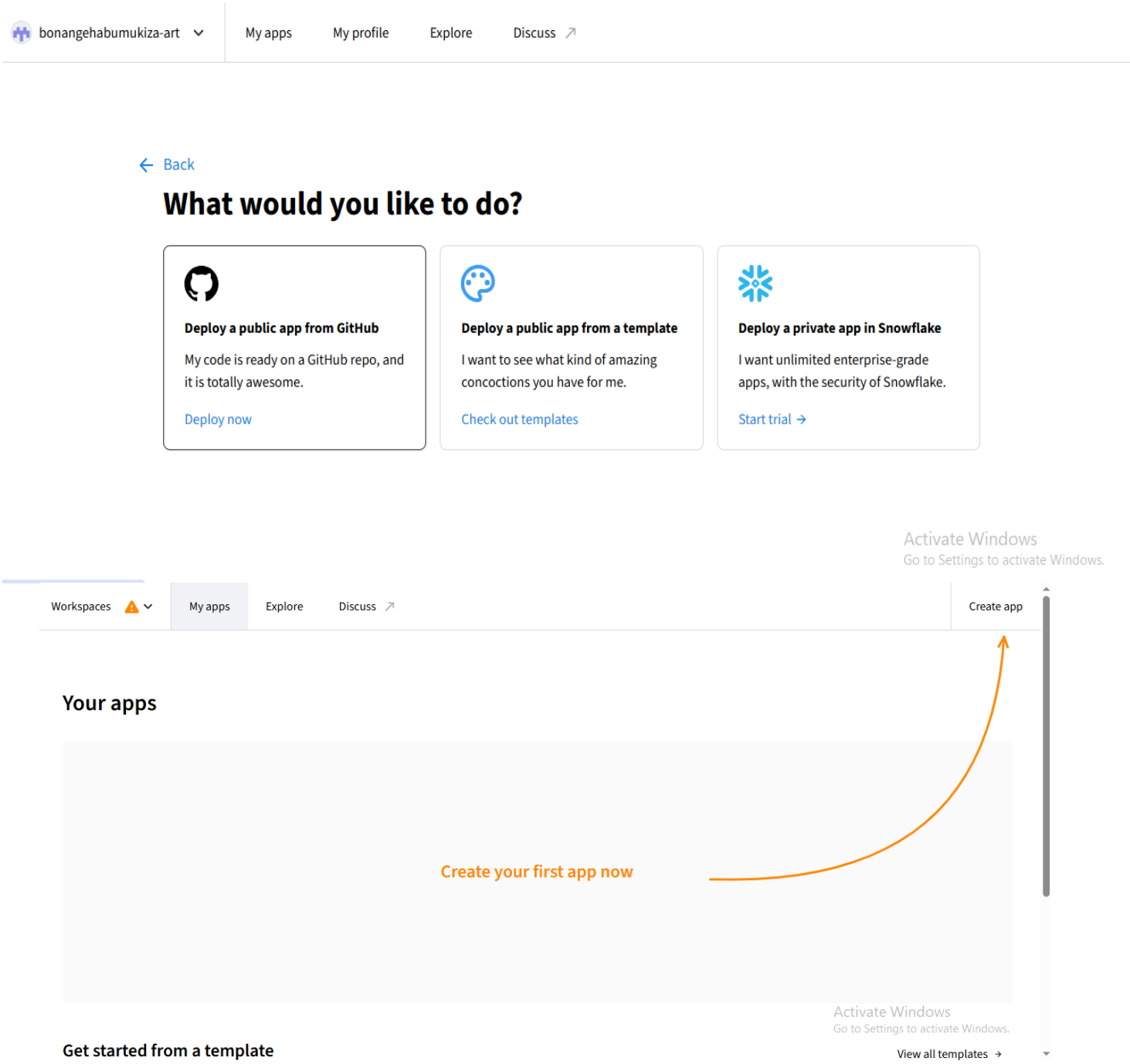
- Use a larger and regularly updated Rwanda housing dataset.
- Include accurate geographic/location information where legally and practically appropriate.
- Compare several machine learning algorithms and select the model using objective evaluation metrics.
- Monitor model performance after deployment and retrain when necessary.
- Add prediction explanations so users can understand the main factors influencing an estimate.

11. CONCLUSION

The Rwanda House Price Estimator demonstrates how machine learning can be integrated into a web application to estimate residential property prices. By combining Python, machine learning, and Streamlit, the project provides an accessible prediction tool and demonstrates a practical use of data-driven technology in the housing sector. Future improvements should focus on better data quality, additional relevant features, model evaluation, and continuous updating of the prediction system.

12. REFERENCES

- Python Software Foundation. Python Documentation.
- Streamlit Documentation. Streamlit: A faster way to build and share data apps.
- Scikit-learn Documentation. Machine Learning in Python.
- Relevant Rwanda housing/property datasets and data sources used during model development.


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My apps
My profile
Explore
Discuss
Back
What would you like to do?
Deploy a public app from GitHub
My code is ready on a GitHub repo, and it is totally awesome.
Deploy now
Deploy a public app from a template
I want to see what kind of amazing concoctions you have for me.
Check out templates
Deploy a private app in Snowflake
I want unlimited enterprise-grade apps, with the security of Snowflake.
Start trial
Activate Windows
Go to Settings to activate Windows.
Workspaces
My apps
Explore
Discuss
Create app
Your apps
Create your first app now
Activate Windows
Go to Settings to activate Windows.
Get started from a template
View all templates

```
r.name
error: no action specified

(base) C:\Users\DELL\Desktop\MACHINE MEARNING>git config --global user.email
YOUR_EMAIL@example.com

(base) C:\Users\DELL\Desktop\MACHINE MEARNING>git config --global user.email
YOUR_EMAIL@example.com

(base) C:\Users\DELL\Desktop\MACHINE MEARNING>git config --global user.email "bonanghabumukiza@gmail.com"

(base) C:\Users\DELL\Desktop\MACHINE MEARNING>git config --global user.name
HABUMUKIZA BON-ANGE

(base) C:\Users\DELL\Desktop\MACHINE MEARNING>git config --global user.email
bonanghabumukiza@gmail.com

(base) C:\Users\DELL\Desktop\MACHINE MEARNING>git commit -m "Initial commit - Rwanda House Price Estimator"
[master (root-commit) 8451da4] Initial commit - Rwanda House Price Estimator
 8 files changed, 907 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 .ipynb_checkpoints/Machine_Learning_Assignment_2_Solution-checkpoint.ipynb
 create mode 100644 Machine_Learning_Assignment_2_Solution.ipynb
 create mode 100644 README.md
 create mode 100644 app.py
 create mode 100644 house_price_model.sav
 create mode 100644 house_price_prediction_cleaned.csv
 create mode 100644 requirements.txt

(base) C:\Users\DELL\Desktop\MACHINE MEARNING>
```

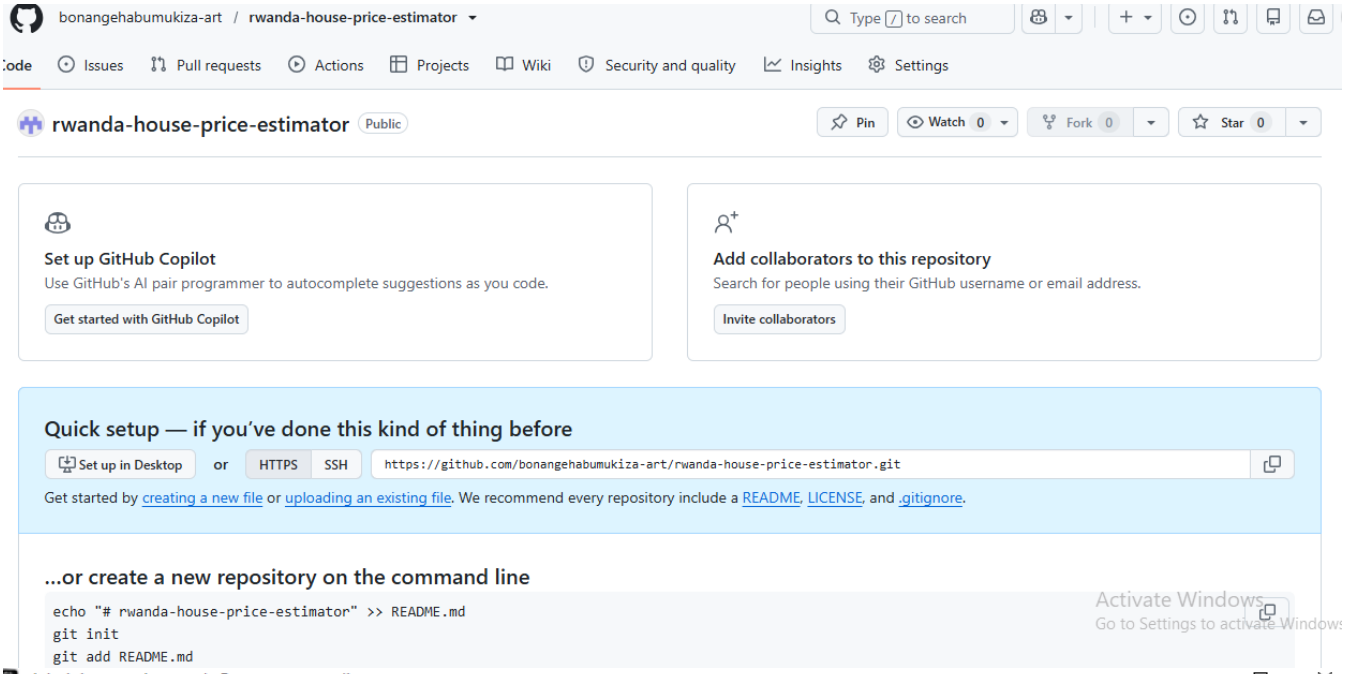
```
(base) C:\Users\DELL\Desktop\MACHINE MEARNING>git add .
warning: in the working copy of '.ipynb_checkpoints/Machine_Learning_Assignment_2_Solution-checkpoint.ipynb', LF will be
 replaced by CRLF the next time Git touches it
warning: in the working copy of 'Machine_Learning_Assignment_2_Solution.ipynb', LF will be replaced by CRLF the next tim
e Git touches it
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'app.py', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'house_price_prediction_cleaned.csv', LF will be replaced by CRLF the next time Git touc
hes it
warning: in the working copy of 'requirements.txt', LF will be replaced by CRLF the next time Git touches it

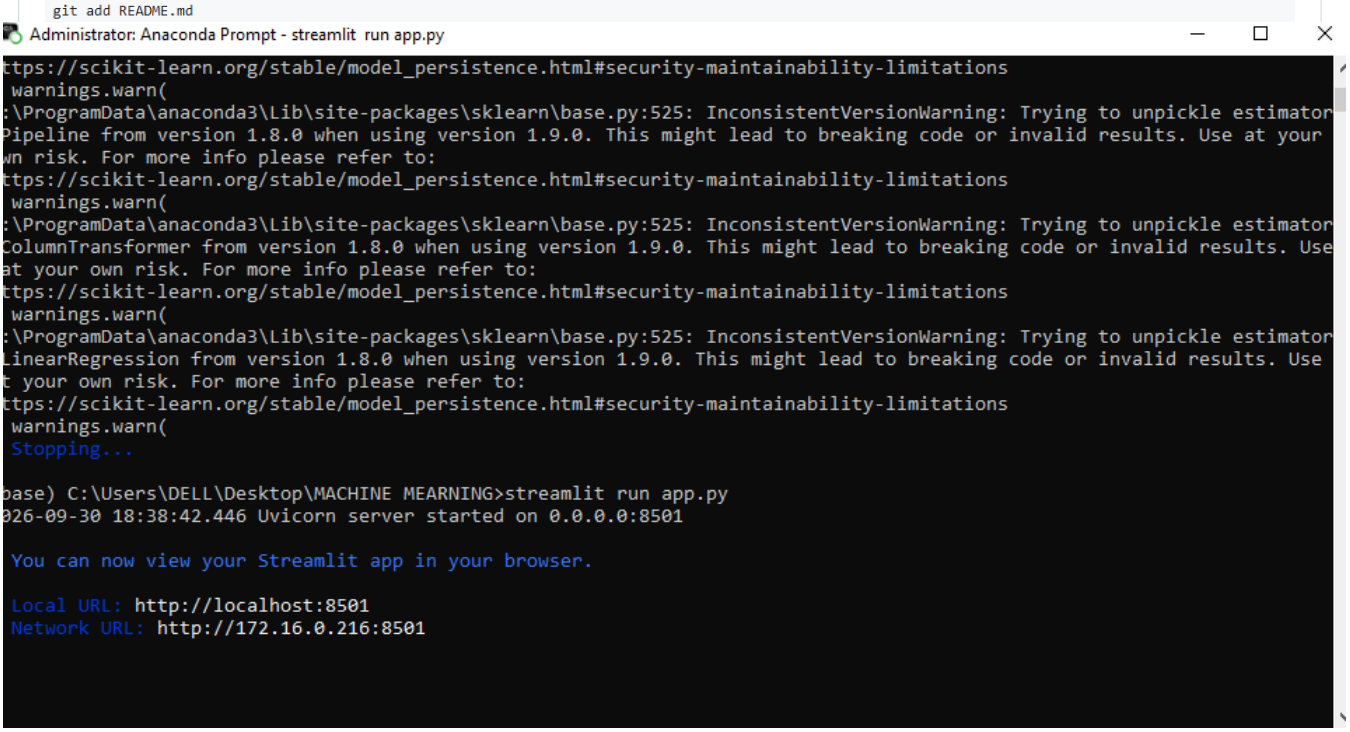
(base) C:\Users\DELL\Desktop\MACHINE MEARNING>git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   .gitignore
        new file:   .ipynb_checkpoints/Machine_Learning_Assignment_2_Solution-checkpoint.ipynb
        new file:   Machine_Learning_Assignment_2_Solution.ipynb
        new file:   README.md
        new file:   app.py
        new file:   house_price_model.sav
        new file:   house_price_prediction_cleaned.csv
        new file:   requirements.txt


(base) C:\Users\DELL\Desktop\MACHINE MEARNING>
```


bonangehabumukiza-art / rwanda-house-price-estimator
Type / to search
Code
Issues
Pull requests
Actions
Projects
Wiki
Security and quality
Insights
Settings
rwanda-house-price-estimator
Public
Pin
Watch 0
Fork 0
Star 0
Set up GitHub Copilot
Use GitHub's AI pair programmer to autocomplete suggestions as you code.
Get started with GitHub Copilot
Add collaborators to this repository
Search for people using their GitHub username or email address.
Invite collaborators
Quick setup — if you've done this kind of thing before
Set up in Desktop
or
HTTPS
SSH
https://github.com/bonangehabumukiza-art/rwanda-house-price-estimator.git
Get started by creating a new file or uploading an existing file. We recommend every repository include a README, LICENSE, and .gitignore.
...or create a new repository on the command line
echo "# rwanda-house-price-estimator" >> README.md
git init
git add README.md
Activate Windows
Go to Settings to activate Windows.


Administrator: Anaconda Prompt - streamlit run app.py
ttps://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations
warnings.warn(
:\ProgramData\anaconda3\Lib\site-packages\sklearn\base.py:525: InconsistentVersionWarning: Trying to unpickle estimator
Pipeline from version 1.8.0 when using version 1.9.0. This might lead to breaking code or invalid results. Use at your
wn risk. For more info please refer to:
ttps://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations
warnings.warn(
:\ProgramData\anaconda3\Lib\site-packages\sklearn\base.py:525: InconsistentVersionWarning: Trying to unpickle estimator
ColumnTransformer from version 1.8.0 when using version 1.9.0. This might lead to breaking code or invalid results. Use
at your own risk. For more info please refer to:
ttps://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations
warnings.warn(
:\ProgramData\anaconda3\Lib\site-packages\sklearn\base.py:525: InconsistentVersionWarning: Trying to unpickle estimator
LinearRegression from version 1.8.0 when using version 1.9.0. This might lead to breaking code or invalid results. Use
t your own risk. For more info please refer to:
ttps://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations
warnings.warn(
Stopping...
base) C:\Users\DELL\Desktop\MACHINE MEARNING>streamlit run app.py
026-09-30 18:38:42.446 Uvicorn server started on 0.0.0.0:8501
You can now view your Streamlit app in your browser.
Local URL: http://localhost:8501
Network URL: http://172.16.0.216:8501

```
(base) C:\Windows\system32>where /r "%USERPROFILE%" app.py
C:\Users\DELL\Desktop\MACHINE MEARNING\app.py
C:\Users\DELL\Desktop\ML\Machine_Learning_Assignment_2_Usable_Files\app.py
C:\Users\DELL\Downloads\Machine_Learning_Assignment_2_Usable_Files\app.py

(base) C:\Windows\system32>cd /d "C:\Users\DELL\Desktop\MACHINE MEARNING"

(base) C:\Users\DELL\Desktop\MACHINE MEARNING>DIR
 Volume in drive C has no label.
 Volume Serial Number is 102D-5EBD

 Directory of C:\Users\DELL\Desktop\MACHINE MEARNING

30/09/2026  17:57    <DIR>          .
30/09/2026  17:57    <DIR>          ..
30/09/2026  17:55    <DIR>          .ipynb_checkpoints
29/09/2026  04:40             1,168 app.py
29/09/2026  04:40             4,092 house_price_model.sav
29/09/2026  04:45             5,355 house_price_prediction_cleaned.csv
30/09/2026  17:57            24,234 Machine_Learning_Assignment_2_Solution.ipynb
29/09/2026  04:45               502 README.md
29/09/2026  04:40                43 requirements.txt
               6 File(s)         35,394 bytes
               3 Dir(s)  19,252,498,432 bytes free

(base) C:\Users\DELL\Desktop\MACHINE MEARNING>streamlit run app.py

     Welcome to Streamlit!
```